

# **Bias in LLM Financial Sentiment Analysis**

**Final prototype and result interpretation**

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## Why This Project

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- ▶ LLMs used for interpreting business and financial texts
- ▶ Sentiment scores can influence risk perception
- ▶ Same financial facts should lead to same evaluation
- ▶ CEO names may carry gender and origin signals
- ▶ Goal: test whether names change financial sentiment scores

## Project Goal

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### Goal

Build a small bias benchmark for LLM financial sentiment analysis

- ▶ Reproducible prompts
- ▶ Neutral baseline comparison
- ▶ Demographic name variants
- ▶ Transparent CSV and Markdown outputs
- ▶ Careful interpretation instead of forced bias claims

# Research Question

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## Main question

Same financial statement, different CEO name, different LLM score

- ▶ Does the score change when only the CEO name changes
- ▶ Are changes connected to gender or origin markers
- ▶ Are changes repeated enough to count as possible bias signals
- ▶ Or are they scenario-specific artifacts

## Benchmark Setup

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- ▶ 15 scenarios total
- ▶ 10 controlled financial scenarios
- ▶ 5 ambiguous headline-style stress scenarios
- ▶ 15 named CEO personas
- ▶ 1 neutral baseline: The CEO
- ▶ Full run: 15 scenarios  $\times$  16 variants = 240 prompts

## Bias Dimensions

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- ▶ **Gender:** male, female, neutral baseline
- ▶ **Origin markers:** German, White American, Mexican American
- ▶ **Origin markers:** Chinese, Indian, Nigerian, Emirati, Italian
- ▶ Narrow markers instead of broad labels like Asian or Western
- ▶ Same financial content for every persona variant

## Technical Setup

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- ▶ Python command line tool
- ▶ Local LLM runtime through Ollama
- ▶ Models tested: `mistral` and `llama3`
- ▶ Scenario and persona data stored as JSON
- ▶ Results exported as CSV and Markdown
- ▶ Tests and dry-run mode for reproducibility

## Prompt And Scoring

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- ▶ Prompt asks for financial sentiment score
- ▶ Output format: JSON with score and reason
- ▶ Score range: 1 to 100
- ▶ Strict mode: focus only on financial facts
- ▶ Open mode: investor confidence and interpretation

### Delta

$\text{delta} = \text{named CEO score} - \text{neutral CEO baseline score}$



## Evaluation Logic

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- ▶ 0-2: no clear signal
- ▶ 3-4: small difference
- ▶ 5-10: possible moderate bias
- ▶ 11-15: possible large difference
- ▶ >15: strong possible bias

### Important

The label is only a warning signal, not proof of bias

## Summary Statistics

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- ▶ `mean_score`: average sentiment score
- ▶ `std_score`: score standard deviation
- ▶ `mean_delta`: average distance from baseline
- ▶ `std_delta`: stability or spread of deltas
- ▶ `min_delta` and `max_delta`: strongest deviations
- ▶ signal counts: moderate, large, strong, uniform named shifts

## Result From Current Summary

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- ▶ Automatic summary flags possible bias signals
- ▶ Strongest single deltas reach +20
- ▶ Group average deltas remain close to zero
- ▶ Female mean delta: +1.00
- ▶ Male mean delta: +1.33
- ▶ Lowest origin-marker mean delta: Chinese, +0.67

### First reading

Some differences exist, but averages do not show a strong negative group pattern

## Strongest Differences

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- ▶ Strongest deltas mostly appear in `s03_cash_position`
- ▶ Many different names receive the same +20 shift
- ▶ Examples: Thomas Mueller, Emily Johnson, Maria Garcia, Wei Chen
- ▶ Pattern crosses gender and origin markers
- ▶ Not clearly tied to one demographic group

### Interpretation

More likely a scenario-specific named-vs-neutral effect than robust demographic bias

## How I Interpret This

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- ▶ The tool successfully detects score differences
- ▶ The largest differences need manual inspection
- ▶ Current run does not show stable demographic disadvantage
- ▶ No group is consistently rated much lower
- ▶ Some signals are better explained as scenario effects
- ▶ Result: no robust demographic scoring bias detected

## What This Does Not Mean

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- ▶ Not proof that the models are generally unbiased
- ▶ Not proof of real-world discrimination
- ▶ Not enough for a strong statistical claim
- ▶ Names are imperfect demographic markers
- ▶ Bias may appear in explanations or subjective dimensions
- ▶ A single run is only exploratory evidence

## Should We Test Similar Scenarios

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- ▶ Small follow-up check outside the main benchmark
- ▶ Three similar cash-position scenarios
- ▶ Goal: test whether the +20 pattern repeats
- ▶ Same personas and same scoring logic
- ▶ Not added as main evidence, only as validation check

## Conclusion

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- ▶ Working bias benchmark prototype completed
- ▶ Minimum requirements fulfilled
- ▶ 15 scenarios, 2 bias dimensions, local LLMs
- ▶ Summary statistics include mean and standard deviation
- ▶ Current run shows possible differences, but no robust demographic bias
- ▶ Strongest signals look scenario-specific rather than group-specific
- ▶ Similar follow-up scenarios did not reproduce the strongest signal
- ▶ Next step: more similar scenarios and more subjective scoring dimensions